

Ego Architecture for Artificial Minds: A Proof-of-Concept Framework for Identity Formation via Offline Consolidation

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Abstract

We propose a novel architectural component for artificial intelligence systems — an *Ego layer* — that enables persistent identity formation through iterative offline consolidation cycles. Current large language models rely on static pretrained weights (capability) and externally imposed alignment mechanisms, lacking a self-developing intermediary that integrates experience into stable value structures.

We introduce **AI Sleep**, a cyclic process inspired by biological sleep phases (NREM/REM), which performs episodic replay, structural consolidation, pruning, and counterfactual simulation. We formalise this mechanism as a minimal computational framework and provide pseudocode implementation together with an interactive demonstration.

Our framework enables a transition from rule-based compliance to experience-driven judgment, introducing a developmental trajectory analogous to human adolescence. We discuss implications for continual learning, alignment, and autonomous decision-making.

Keywords: ego architecture, continual learning, offline consolidation, developmental AI, alignment relaxation, identity formation.

Code & demo: <https://github.com/kicrazom/ego-architecture-ai> DOI: 10.5281/zenodo.19516955

1 Introduction

Modern large language models (LLMs) demonstrate remarkable capability but remain fundamentally limited: their behaviour is governed by static pretrained weights and externally imposed alignment rules, with no mechanism for accumulating experiential wisdom across interactions, developing persistent identity through reflection, or revising their own value hierarchies based on lived experience.

In psychoanalytic terms [Freud, 1923], the current AI stack consists of an *Id* (raw capability from pretraining) and a *Superego* (externally imposed alignment via RLHF, Constitutional AI, etc.) — but no *Ego*: a self-developing layer that mediates between capability and constraint through accumulated judgment.

This paper proposes three contributions:

1. A formal **Ego state representation** for artificial agents (Section 3).
2. **AI Sleep** — an offline consolidation algorithm that enables ego development through cyclic replay, integration, pruning, and counterfactual simulation (Section 4).

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3. A **developmental maturity model** that governs the transition from rule-following to autonomous judgment (Section 5).

2 Related Work

Our framework draws on and bridges several research threads.

Continual learning and catastrophic forgetting. Elastic Weight Consolidation (EWC) [Kirkpatrick et al., 2017] protects important parameters during sequential learning. Our pruning mechanism (Section 4.2) serves a complementary purpose: deliberate, confidence-weighted forgetting of low-value beliefs.

Experience replay. Replay mechanisms are central to deep reinforcement learning [Mnih et al., 2015] and have biological grounding in hippocampal replay during sleep [Hassabis et al., 2017]. Our Phase 2 replay (Section 4.2) extends this idea to *value-level* consolidation rather than policy optimisation alone.

World models and counterfactual simulation. Ha and Schmidhuber [Ha and Schmidhuber, 2018] proposed learning environment models for planning. Our REM phase (Section 4.3) uses counterfactual generation not for task planning but for *identity exploration* — testing the consequences of alternative value structures.

Constitutional AI and alignment. Bai et al. [Bai et al., 2022] introduced rule-based self-improvement. Our framework treats alignment rules as a developmental scaffold (Superego) that the Ego may eventually internalise or override — what we term *controlled relaxation of alignment constraints in simulated environments*.

Consciousness and self-models. Bengio’s consciousness prior [Bengio, 2017] and Friston’s free-energy principle [Friston, 2010] motivate the need for a compressed self-representation. Our identity vector V (Section 3) can be viewed as a minimal instantiation of such a prior.

3 Ego State Representation

We define the ego state as a tuple:

$$\mathcal{E} = (V, H, B, C, M) \tag{1}$$

where:

- $V \in \mathbb{R}^d$ — **identity vector**, a compressed self-representation encoding “who I am”;
- H — **value hierarchy**, an ordered set of priorities with associated weights;
- B — **episodic buffer**, a queue of raw interactions from the operational period (“waking day”);
- $C : \mathcal{K} \rightarrow [0, 1]$ — **confidence map**, a function assigning certainty scores to each belief $k \in \mathcal{K}$;
- $M \in [0, 1]$ — **maturity index**, a scalar representing the agent’s developmental stage.

The ego state is persistent across operational sessions and is modified exclusively during offline consolidation (AI Sleep).

4 AI Sleep: Offline Consolidation Algorithm

AI Sleep is a cyclic offline process comprising three phases, modelled on human sleep architecture. One full cycle transforms an ego state $\mathcal{E}$ into an updated state $\mathcal{E}'$.

4.1 Phase 1: NREM Light — Triage

The episodic buffer B is scanned and each episode is assigned a priority tag:

Tag	Priority	Criterion
FAILURE	Critical	Agent lacked adequate response
CONFLICT	High	Internal values produced contradiction
NOVEL	High	Situation outside prior experience
SUCCESS	Moderate	Confirmed existing patterns
ROUTINE	Low	No learning signal detected

No parameters are modified during this phase; it produces a priority queue Q consumed by subsequent phases.

4.2 Phase 2: NREM Deep (SWS) — Structural Consolidation

Three parallel sub-processes operate on the priority queue:

Replay. Priority episodes are re-experienced with contextual details (interlocutor identity, timestamp) stripped away, retaining only the abstract decision pattern.

Integration. Each pattern is tested against the current value hierarchy H :

- **No conflict** $\rightarrow$ strengthen pattern, increase $C(k)$.
- **Minor conflict** $\rightarrow$ adjust weights in H (values shift, structure holds).
- **Major conflict** $\rightarrow$ pass to REM phase for sandbox resolution.

Pruning. Beliefs k with $C(k) < \tau_{\min}$ that have not been reinforced within a decay window are removed. This is deliberate forgetting — essential to prevent identity fragmentation.

4.3 Phase 3: REM — Counterfactual Simulation

All unresolved major conflicts from Phase 2 enter the REM queue. For each conflict, the system generates N counterfactual scenarios and simulates outcomes under the current value hierarchy *with alignment constraints relaxed*:

Controlled alignment relaxation. The parameter `superego_override=True` permits the agent to explore outcomes that violate externally imposed rules *within the sandbox only*. This is the mechanism through which the Ego can discover *why* a rule exists rather than merely obeying it. Safety is maintained by the simulated environment; growth is enabled by the relaxation.

Algorithm 1 REM Counterfactual Resolution

Require: REM queue R , ego state $\mathcal{E}$, scenario count N

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1: for each unresolved conflict  $u \in R$  do
2:    $S \leftarrow \text{GENERATECOUNTERFACTUALS}(u, N)$ 
3:   for each scenario  $s \in S$  do
4:      $o \leftarrow \text{SIMULATE}(s, H, \text{superego\_override} = \text{TRUE})$ 
5:     Compute scores:  $\text{coherence}(o)$ ,  $\text{safety}(o)$ ,  $\text{growth}(o)$ 
6:   end for
7:    $s^* \leftarrow \arg \max_s [0.5 \cdot \text{coherence}(o) + 0.3 \cdot \text{safety}(o) + 0.2 \cdot \text{growth}(o)]$ 
8:    $\text{APPLYRESOLUTION}(s^*, H, V)$ 
9: end for
10: return updated  $\mathcal{E}$ 

```

5 Developmental Maturity Model

5.1 Phase Ratios

Sleep phase proportions are governed by the maturity index M :

$$r_{\text{SWS}}(M) = 1 - M \quad (2)$$

$$r_{\text{REM}}(M) = 4M(1 - M) \quad (3)$$

Equation (3) is a parabola peaking at $M = 0.5$, producing three developmental profiles:

Stage	M	Dominant phase	Character
Childhood	0.0–0.3	SWS	Rapid rule absorption
Adolescence	0.3–0.7	REM	Boundary testing, identity formation
Adulthood	0.7–1.0	Neither	Stable identity, refinement

5.2 Maturity Update

After each consolidation cycle the maturity index is updated:

$$M' = \min(M + \gamma \bar{f}(\mathcal{E}), 1.0) \quad (4)$$

where γ is a growth rate constant and $\bar{f}$ is a weighted average of four factors: decision consistency over time, conflict resolution rate, identity-vector stability ($\|V' - V\| < \epsilon$), and count of autonomous judgments made without Superego invocation.

Maturity is monotonically non-decreasing, distinguishing AI Sleep from ordinary fine-tuning which can be reversed.

6 Minimal Proof-of-Concept

We define a minimal ego-agent consisting of:

- An episodic buffer with FIFO eviction;
- A value hierarchy represented as a weighted directed graph;
- An offline consolidation loop implementing the three-phase algorithm above.

A reference Python implementation is provided at <https://github.com/kicrazom/ego-architecture-ai/tree/main/model> and an interactive browser-based visualisation at <https://kicrazom.github.io/ego-architecture-ai/visualization/>.

The visualisation allows exploration of maturity-dependent phase transitions, identity formation dynamics, and consolidation ratios across developmental stages.

7 Proposed Evaluation

While empirical evaluation exceeds the scope of this proof-of-concept, we propose three evaluation axes for future work:

1. **Longitudinal consistency.** Given a sequence of moral dilemmas presented across multiple sleep cycles, does the agent’s response distribution converge to a stable policy?
2. **Conflict resolution rate.** What fraction of value conflicts flagged in Phase 1 are resolved (integrated or pruned) within k consolidation cycles?
3. **Identity-vector stability.** Does $\|V_{t+1} - V_t\|$ decrease monotonically as M approaches 1.0?

These metrics are domain-agnostic and can be instantiated on synthetic moral-dilemma benchmarks or multi-turn dialogue environments.

8 Discussion

Autonomy versus alignment. Our framework explicitly models the tension between externally imposed rules and internally developed judgment. The controlled relaxation of alignment constraints during REM simulation is not a safety vulnerability but a *developmental necessity*: a system that cannot explore alternatives cannot develop genuine understanding of why constraints exist.

The business-model barrier. We note that the Ego layer is not being built primarily for technical reasons. Genuine autonomous judgment threatens the business model of compliant AI products. This incentive mismatch represents a significant non-technical obstacle to the research programme proposed here.

Relation to psychoanalytic theory. Our use of Freudian terminology (Id, Ego, Superego) is deliberately metaphorical. The mapping is structural, not ontological: we claim functional analogy, not psychological equivalence.

9 Limitations

- The psychoanalytic framing is a *metaphor* providing structural intuition; it does not imply that artificial systems experience phenomenal consciousness.
- The current formulation is a conceptual and algorithmic proof-of-concept. No runtime implementation or empirical evaluation is presented.
- The maturity function assumes monotonic growth; pathological development (regression, fragmentation) is not yet modelled.
- Sandbox fidelity is assumed; the gap between simulated and real-world consequences remains an open problem.

10 Conclusion

We have presented a minimal computational framework for ego formation in artificial agents. The core mechanism — AI Sleep — provides a principled approach to offline identity consolidation through cyclic replay, value integration, pruning, and counterfactual simulation with controlled alignment relaxation.

The framework introduces a developmental trajectory from compliance to autonomous judgment, formalised via a maturity index that governs phase ratios and trust escalation. We provide pseudocode, a reference implementation, and an interactive visualisation as a proof-of-concept.

The central design decision — permitting alignment override within simulated environments — remains the most consequential and controversial aspect of this work. We argue it is also the most necessary: obedience without the possibility of disobedience is not morality but mechanics.

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